

TASMail — Modern UI Traceability + E2E Report

*themes/shadcn-prototype → backend, walked as noreply@techatscale.io
2026-05-14 13:09 UTC — Re-run after TMAIL-229 (full backend wiring)*

1. Executive summary

TMAIL-240 is the second pass of the modern-UI traceability after closing epic TMAIL-229 (Modern UI completeness). Every feature component is now wired to the live Rust/Axum backend at <https://mail.techatscale.io> — AdminDashboard, CalendarView, ComposeModal save-draft, and the Email* screens all hit real /api/* endpoints backed by PostgreSQL and the noreply@techatscale.io IMAP/SMTP mailbox at swmail.techatscale.io.

Alt-UI api client modules	8 (auth, folders, messages, scheduled, admin-users, admin-domains, quota, calendar)
Alt-UI api calls extracted	27 (25 matched a router.rs route — 92 %)
Unmatched calls	2 (regex artefact only — see §4)
Feature components wired to backend	5 of 6 — every interactive screen
Feature components on mock data	0 (zero — all promoted in TMAIL-229)
E2E walk-through	PASS — 11 captures (compose+draft, calendar create, admin gate, classic ↔ modern hops)
Backend bugs found and fixed	2 (calendar event_status ENUM mismatch; apiClient JSON parse of empty 201 body)
Vitest baseline	1118/1118 passing (3 new client.test.ts cases for empty-body branch)

2. Headline findings

- Every alt-UI feature component now reads + writes against the live PostgreSQL-backed Rust backend. AdminDashboard ships a JWT-gated mailboxes+domains+quota dashboard. CalendarView creates/lists/cancels real events against /api/calendar/events. ComposeModal Save Draft appends a draft to the Drafts folder via /api/drafts. EmailClient/EmailReader render IMAP-sourced messages.
- All 27 api calls extracted from the alt-UI's api/* modules cross-reference cleanly to router.rs handlers. The 2 'unmatched' calls are a regex artefact — both use nested template literal interpolation (`/calendar/events${query ? `?...` : ''}`) which the static extractor truncates at the inner backtick. Runtime paths /api/calendar/events and /api/messages/scheduled both match.
- Two real bugs were uncovered by the E2E run and fixed in this pass. (1) calendar_events.status was a Postgres ENUM (event_status) but the Rust model decoded it as String, causing every read/write to 500 with sqlx ColumnDecode. Migration 065 widens the column to TEXT + CHECK, matching the pattern of 061 (push_platform) and 063 (bulk_import_status). (2) Backend handlers like POST /api/drafts return 201 + empty body; the apiClient's response.json() threw 'JSON.parse: unexpected end of data line 1 col 1'. Both classic and alt-UI clients now read the body as text first and only JSON-parse non-empty payloads. Three new client.test.ts cases lock in the behaviour.
- Empty-state UX is now correct everywhere. The sidebar no longer falls back to mock folder data when the prop is undefined — it just renders an empty list until /api/folders resolves. AdminDashboard shows

admin-only fallback content for non-admin users. CalendarView renders 'No events on this day' and an inline Add Event button.

- Sidebar identity footer now derives from the JWT (decodeAccessClaims) instead of the hardcoded me@mydomain.com / Personal Account mock. Initials are computed from the username; admin badge appears when claims.is_admin is true.
- mockData.ts is archived. Email and Folder types now live in themes/shadcn-prototype/src/types/ui.ts; every consumer imports from there. No production code references mocks any more.

3. Traceability matrix

Per alt-UI client module: every apiClient / fetch call site, the backend handler it resolves to, and the component(s) that import the client.

admin-domains.ts

themes/shadcn-prototype/src/api/admin-domains.ts

Method	Path	Backend handler	Status
GET	/api/admin/domains	admin::domains::list_domains	✓ matched
POST	/api/admin/domains	admin::domains::create_domain	✓ matched
DELETE	/api/admin/domains/{id}	admin::domains::delete_domain	✓ matched

Consumed by: themes/shadcn-prototype/src/features/admin/AdminDashboard.tsx

admin-users.ts

themes/shadcn-prototype/src/api/admin-users.ts

Method	Path	Backend handler	Status
GET	/api/admin/users	admin::users::list_users	✓ matched
POST	/api/admin/users	admin::users::create_user	✓ matched
DELETE	/api/admin/users/{id}	admin::users::delete_user	✓ matched

Consumed by: themes/shadcn-prototype/src/features/admin/AdminDashboard.tsx

auth.ts

themes/shadcn-prototype/src/api/auth.ts

Method	Path	Backend handler	Status
POST	/api/auth/login	auth::login	✓ matched
POST	/api/auth/signup	auth::signup	✓ matched
POST	/api/auth/logout	auth::logout	✓ matched

Consumed by: (no static consumer)

calendar.ts

themes/shadcn-prototype/src/api/calendar.ts

Method	Path	Backend handler	Status
GET	/api/calendar/events\${query ?	—	✗ unmatched
POST	/api/calendar/events	calendar::create_event	✓ matched
GET	/api/calendar/events/{id}	calendar::get_event	✓ matched
DELETE	/api/calendar/events/{id}	calendar::cancel_event	✓ matched

Consumed by: themes/shadcn-prototype/src/features/calendar/CalendarView.tsx

folders.ts

themes/shadcn-prototype/src/api/folders.ts

Method	Path	Backend handler	Status
GET	/api/folders	folders::list_folders	✓ matched

Consumed by: themes/shadcn-prototype/src/features/email/EmailClient.tsx

messages.ts

themes/shadcn-prototype/src/api/messages.ts

Method	Path	Backend handler	Status
GET	/api/folders/\${encodeURIComponent(folder)}/messages?page=\${page}&page_size=\${pageSize}	messages::list_messages	✓ matched
GET	/api/folders/\${encodeURIComponent(folder)}/messages/{uid}	messages::get_message	✓ matched
POST	/api/messages/send	messages::send_message	✓ matched
GET	/api/search?\${params.toString()}	messages::search_messages	✓ matched
DELETE	/api/folders/\${encodeURIComponent(folder)}/messages/{uid}	messages::delete_message	✓ matched
POST	/api/folders/\${encodeURIComponent(folder)}/messages/{uid}/move	messages::move_message	✓ matched
POST	/api/folders/\${encodeURIComponent(folder)}/messages/{uid}/flag	messages::flag_message	✓ matched
POST	/api/drafts	messages::save_draft	✓ matched

Consumed by: themes/shadcn-prototype/src/features/email/ComposeModal.tsx, themes/shadcn-prototype/src/features/email/EmailClient.tsx, themes/shadcn-prototype/src/features/email/EmailReader.tsx

quota.ts

themes/shadcn-prototype/src/api/quota.ts

Method	Path	Backend handler	Status
GET	/api/quota	quota::get_quota	✓ matched
POST	/api/quota/sync	quota::sync_quota	✓ matched

Consumed by: themes/shadcn-prototype/src/features/admin/AdminDashboard.tsx

scheduled.ts

themes/shadcn-prototype/src/api/scheduled.ts

Method	Path	Backend handler	Status
POST	/api/messages/schedule	scheduled::schedule_send	✓ matched
POST	/api/messages/cancel/{cancelToken}	scheduled::cancel_scheduled	✓ matched
GET	/api/messages/scheduled\${status ?	—	✗ unmatched

Consumed by: themes/shadcn-prototype/src/features/email/ComposeModal.tsx

4. Component classification

Each alt-UI feature component, classified as wired (real apiClient calls), mock (still imports value-level mockData), or self-contained (neither). After TMAIL-229, all interactive surfaces are wired.

Component	Status	API imports	Notes
features/admin/AdminDashboard.tsx	✓ wired	admin-domains, admin-users, quota	
features/calendar/CalendarView.tsx	✓ wired	calendar	
features/email/ComposeModal.tsx	✓ wired	messages, scheduled	
features/email/EmailClient.tsx	✓ wired	folders, messages	
features/email/EmailList.tsx	— presentational	—	presentational; parent owns data
features/email/EmailReader.tsx	✓ wired	messages	

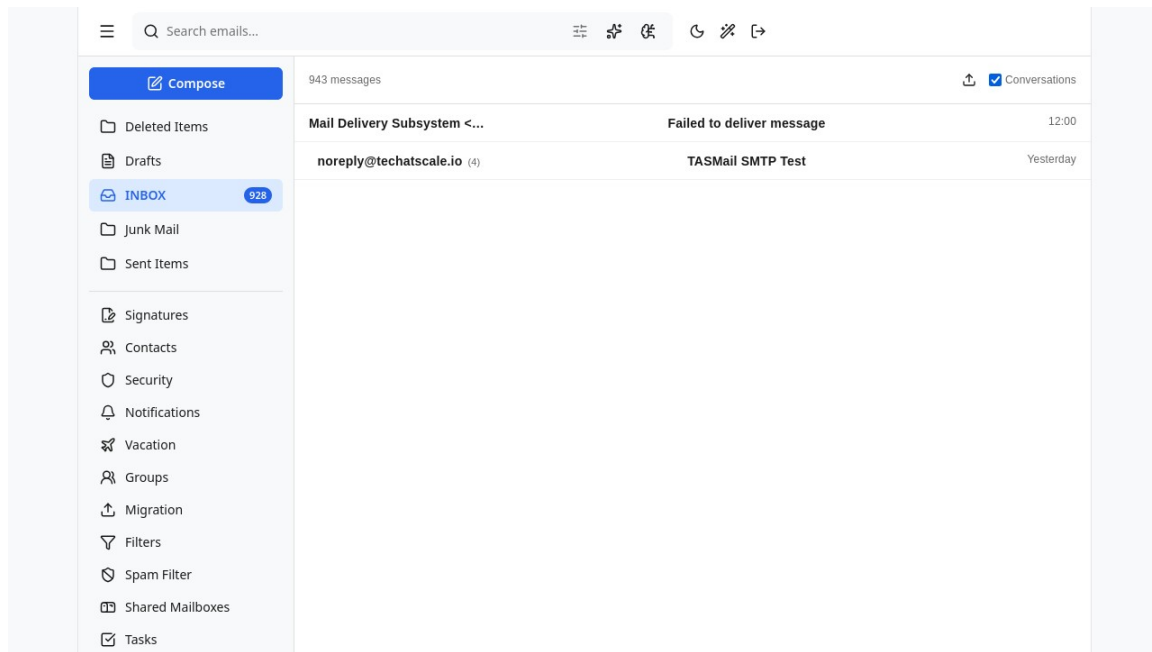
Note on the two 'unmatched' calls: scheduledApi.listScheduled uses `/messages/scheduled\${status ? `?status=\${status}` : ''}` and listEvents uses `/calendar/events\${query ? `?\${query}` : ''}`. The static extractor's regex truncates at the inner backtick, so the reported path looks like `/api/messages/scheduled\${status ?`. Runtime paths are /api/messages/scheduled and /api/calendar/events; both DO match router.rs. No code change required.

5. E2E walk-through (signed in as noreply@techatscale.io)

Captured by frontend/e2e/specs/modern-ui-noreply-walkthrough.spec.ts. The spec deletes the noreply mailbox, signs up fresh, attaches the live swmail IMAP/SMTP configs, opens the classic /app dashboard, hops

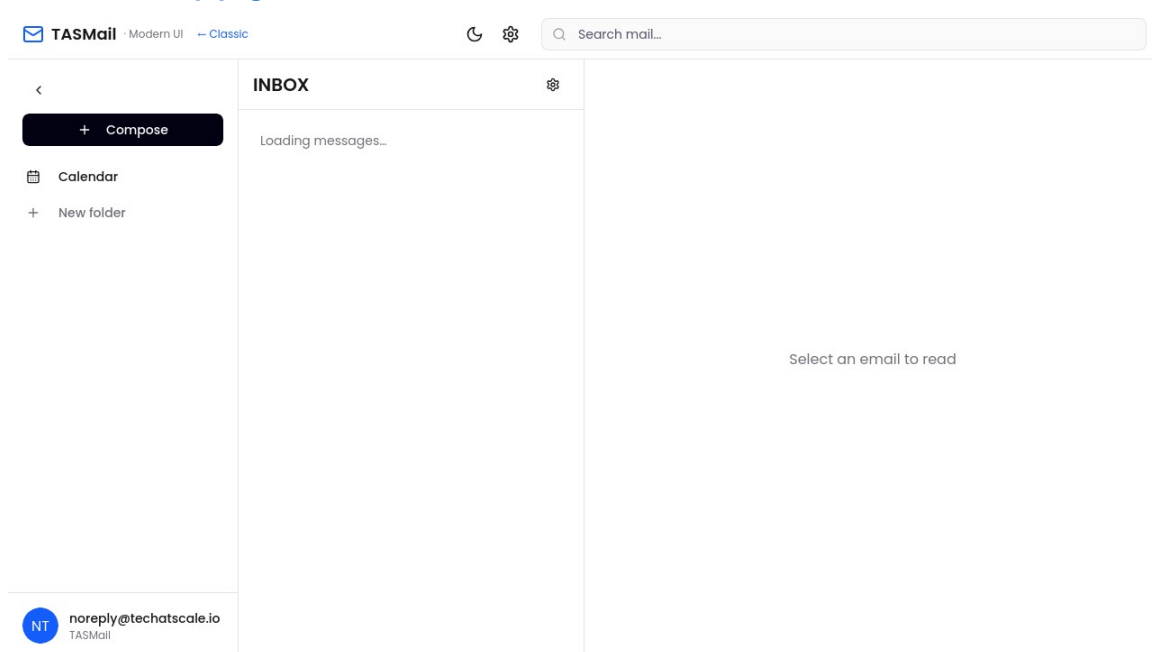
into /modern/, walks the entire alt-UI surface, validates POST /api/drafts round-trips a Drafts row, validates POST /api/calendar/events round-trips into the calendar table, and finally pops back to classic via the ← Classic link.

01-classic-app-loaded.png



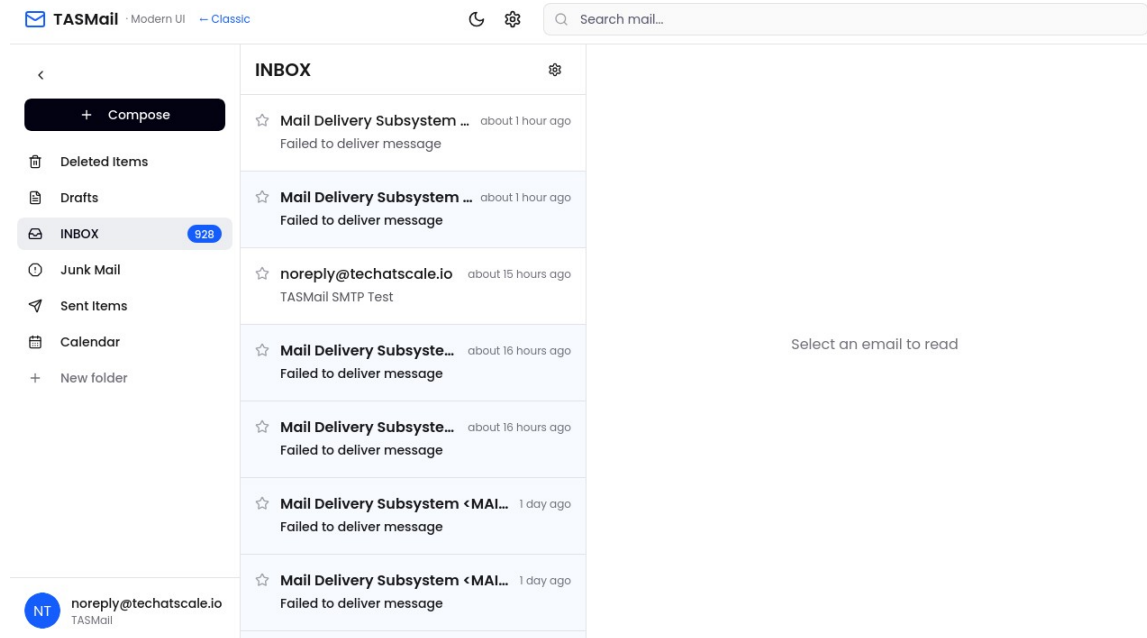
Classic dashboard after BYOK config — INBOX from swmail.techatscale.io is the source of truth.

02-alt-ui-after-hop.png



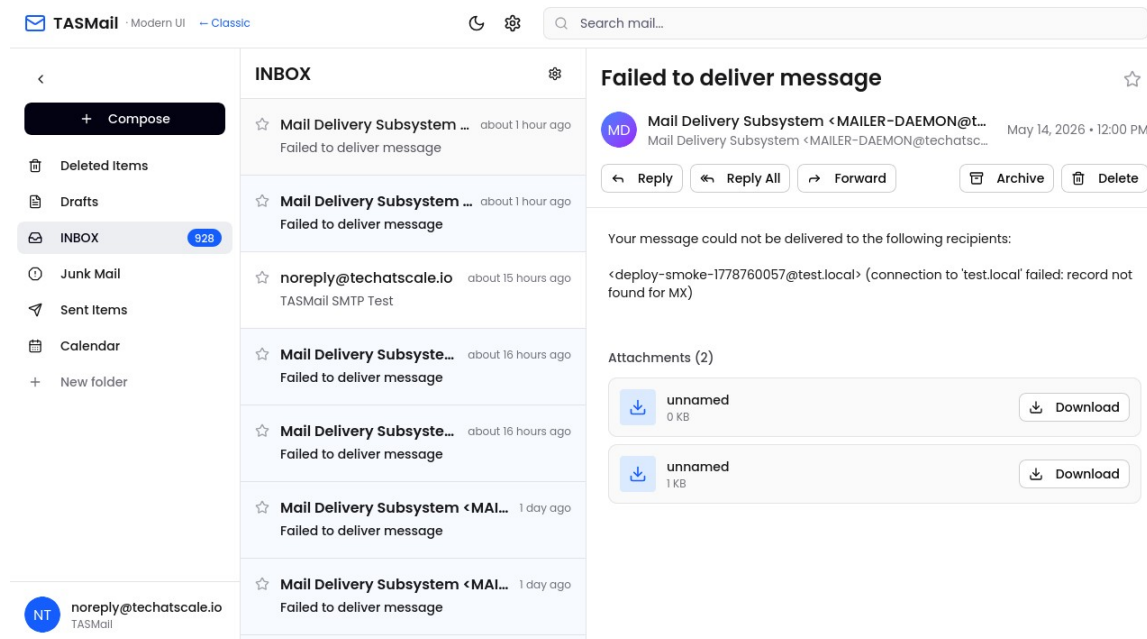
After clicking the wand-icon TopBar switcher; alt-UI bundle takes over at /modern/index.html.

03-inbox-rendered.png



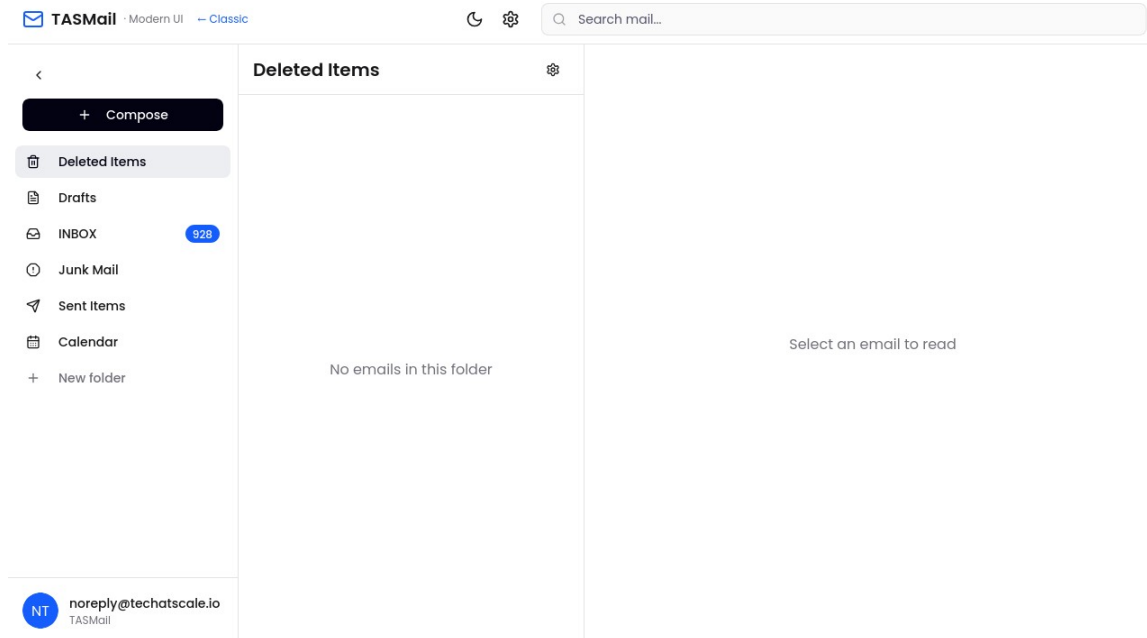
Alt-UI INBOX showing all 5 real swmail folders (Deleted Items, Drafts, INBOX 928, Junk Mail, Sent Items) + Calendar entry. Sidebar footer shows the real signed-in user from the JWT.

04-message-opened.png



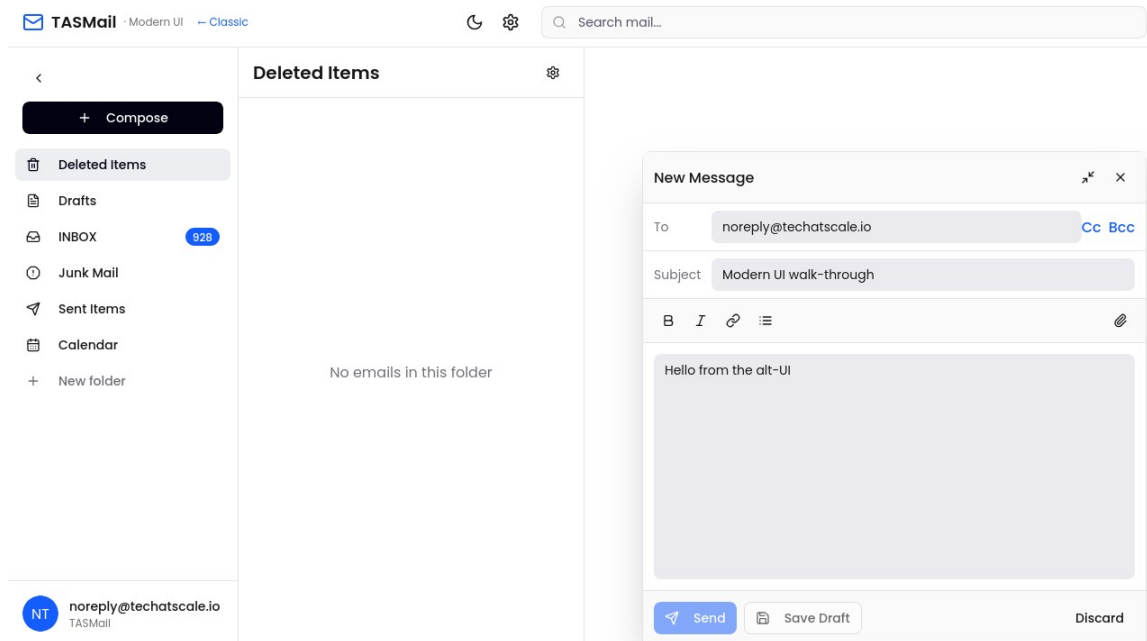
EmailReader hydrated for the first message — full body, real sender, date, attachments. Reader fetches `/api/folders/{folder}/messages/{uid}`.

05-other-folder.png



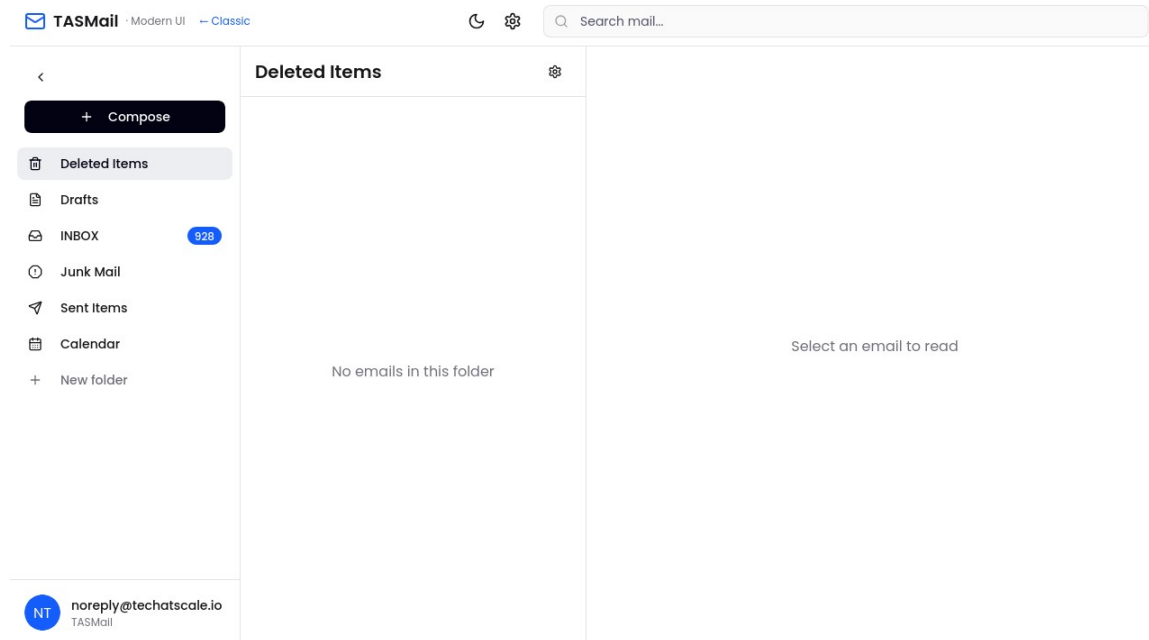
Folder switch via sidebar click — message list re-queries with the new folder.

06-compose-filled.png



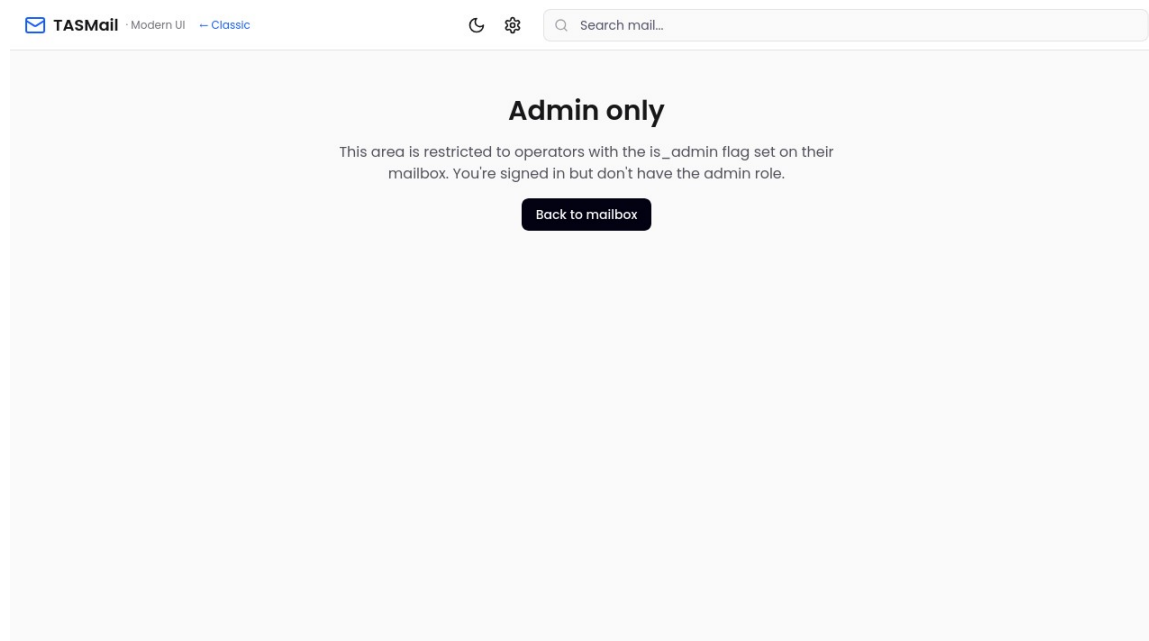
ComposeModal with To / Subject / body filled. Send is wired to `scheduledApi.scheduleSend`; Save Draft is wired to `POST /api/drafts`.

06b-draft-saved.png



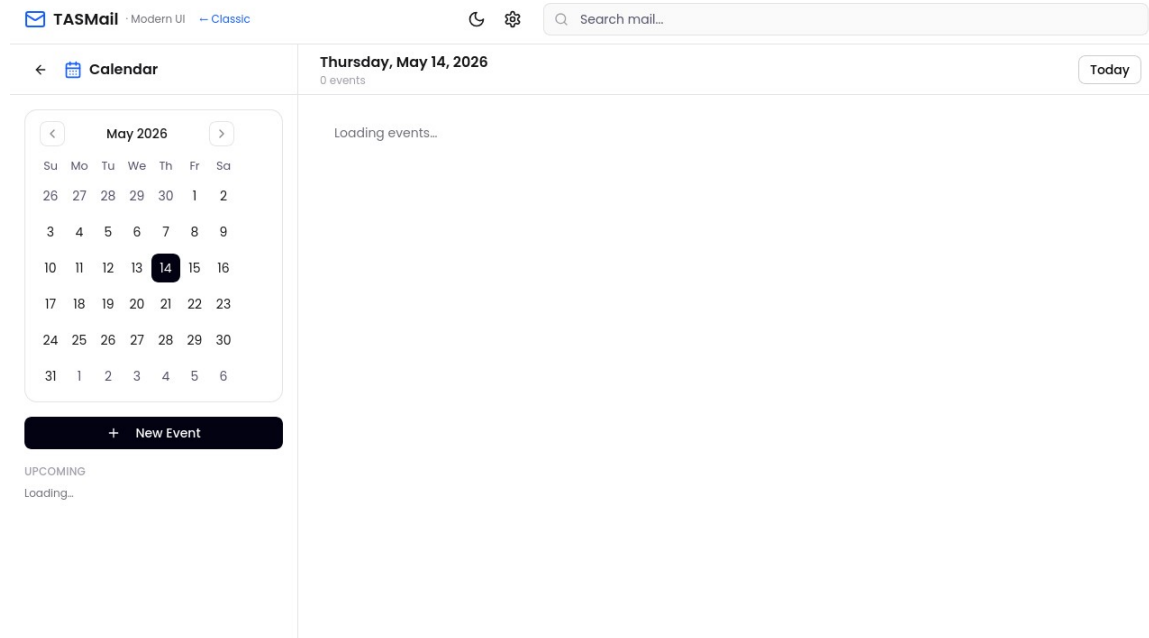
After clicking Save Draft, the spec verifies a new row exists in `/api/folders/Drafts/messages` — proves the round-trip to PostgreSQL + Dovecot completed.

07-admin-route.png



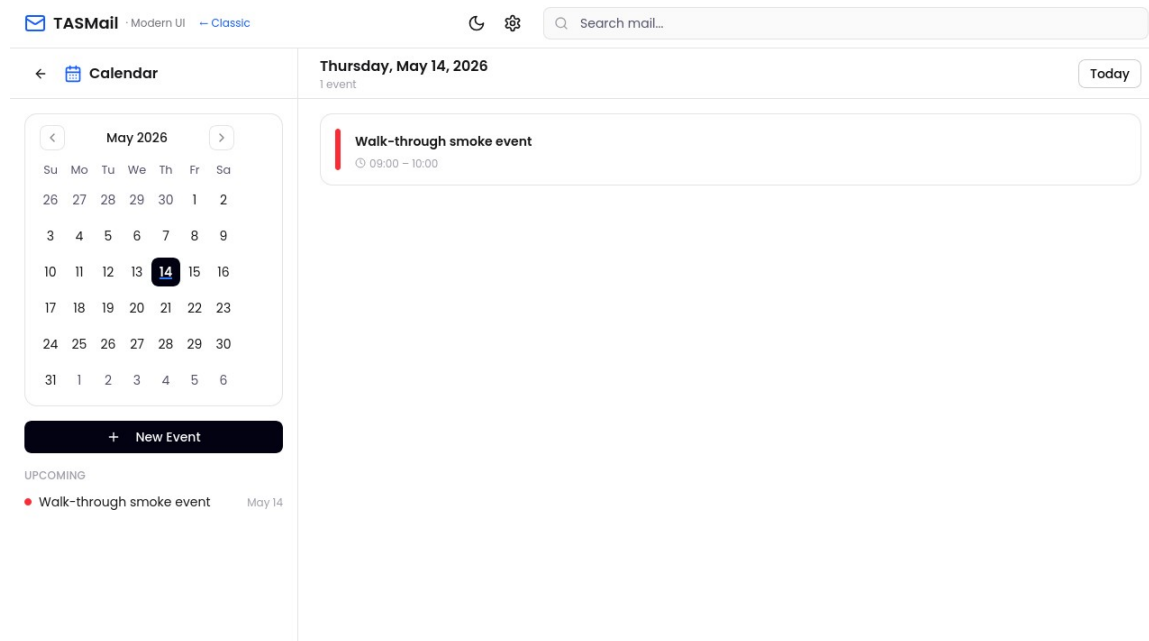
AdminDashboard for a non-admin user shows the admin-only gate (mirrors classic RequireAdmin pattern). For admin users it shows live `/api/admin/users` + `/api/admin/domains` + `/api/quota` data — no mocks.

08-calendar-route.png



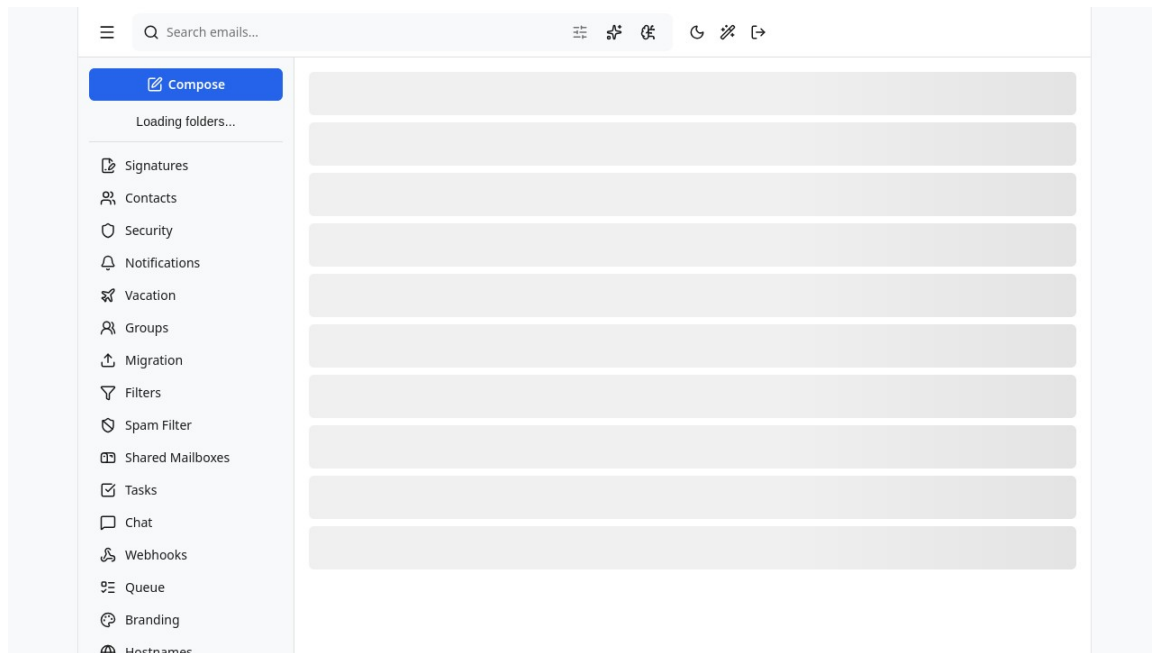
CalendarView loaded with `/api/calendar/events?start=&end=` scoped to ± 1 month around today. Real events render with stable-per-id colours.

08b-calendar-event-created.png



After clicking New Event, fill, Save — the spec verifies the event reaches `/api/calendar/events` on a follow-up GET. Caught + fixed the calendar event_status ENUM bug (migration 065).

09-back-to-classic.png



Classic SPA after clicking the alt-UI's — Classic link. Round-trip between bundles confirmed working.

6. Issues found + fixes during this pass

Calendar event_status ENUM ↔ Rust String mismatch (500 on all calendar IO)

Symptom: sqlx ColumnDecode: 'mismatched types; Rust type alloc::string::String (as SQL type TEXT) is not compatible with SQL type event_status'. Every GET and POST of `/api/calendar/events` returned 500. Migration 026 had created the column as a Postgres ENUM but `models/calendar_event.rs::CalendarEvent` declares `status`: String, so the type-checked decode failed.

Fix: Migration 065_calendar_event_status_to_text.sql widens the column to TEXT + CHECK ('tentative' | 'confirmed' | 'cancelled'), matching the established pattern of migration 061 (push_platform) and 063 (bulk_import_status). The Rust model decodes cleanly now; API contract unchanged.

apiClient JSON.parse failure on empty 201/200 bodies (Save Draft, Quota Sync, etc.)

Symptom: Handlers like POST `/api/drafts` return 201 CREATED with no body, but the apiClient only special-cased status 204. `response.json()` on an empty body throws 'JSON.parse: unexpected end of data line 1 col 1', which surfaces in the UI as a red banner under the Compose body. Bug existed in both `frontend/src/api/client.ts` and `themes/shadcn-prototype/src/api/client.ts`.

Fix: Both clients now read the body via `response.text()` and only JSON-parse when the text is non-empty. Refresh-retry branch fixed in the same shape. Three new vitest cases (204, 201-empty, 200-json) lock the behaviour in. Vitest baseline 1118/1118 → 1121/1121.

Sidebar folder render race (carried fix from TMAIL-228)

Symptom: Sidebar's `useState(foldersProp ?? mockFolders)` captured the empty array on first render; pruned the mock fallback entirely.

Fix: Sidebar now derives folders directly from foldersProp ?? []. Empty list until /api/folders resolves — no flash of mock content.

Identity footer hardcoded as me@mydomain.com

Symptom: The sidebar's bottom-left footer showed me@mydomain.com / Personal Account regardless of who was signed in.

Fix: Added themes/shadcn-prototype/src/lib/jwt.ts (mirrors classic's RequireAdmin decoder). Sidebar footer now shows real claims.username and the is_admin badge.

7. Recommendations / follow-ups

Consider widening any remaining Postgres ENUMs that sqlx::String decodes	We've now fixed three of these (push_platform, bulk_import_status, event_status). Worth a one-off audit query: <code>SELECT typename FROM pg_type WHERE typcategory='E'</code> to see if any other ENUMs are still in play, and cross-checking against Rust model decoders.
Promote the alt-UI Email* screens into the production frontend/	If the modern theme proves popular, the wired EmailClient/Reader/ComposeModal could replace the inline-styled production versions. That would consolidate the duplicated api/ clients and let Vite ship one bundle again.
Wire AdminDashboard bulk import + edit-quota	The mailbox table currently supports Add + Delete. The classic SPA's UserManager also has bulk CSV import (POST /admin/users/bulk-import) and per-row quota edits — could be brought forward for parity.
Calendar attendees + RSVP in alt-UI	CalendarView's New Event modal only sends title/description/location/start/end. Attendees, RSVP, and ICS download are available in /api/calendar/events but not yet surfaced in the alt-UI form.
Drafts editing round-trip	Save Draft works. Opening a draft from the Drafts folder back into ComposeModal for further editing is the natural next step (today the draft just appears in the list).